SU ZIJIN

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EDUCATION

University College London (UCL) London, UK
Master of Science in Physics Expected Graduation July 2026
First Class Honours, Expected
Master's thesis: Deep Learning for Radar Altimetry Echo Classification and Sea Surface Image Recognition
Advisors: *Prof. Ryan Nichol & Dr. Michel Tsamados*
Relevant Coursework: High Level Statistical Analysis of Data(81/100), Techniques of High-Performance Computing(90/100), Physics of Exoplanets, Quantum Computation and Communication, Advanced Quantum Theory

University College London (UCL) London, UK
Bachelor of Science in Physics Graduated June 2025
Second Class Honours, Upper Division
Relevant Coursework: Data Analysis and Statistics, Mathematical Methods Ⅰ&Ⅱ, Practical Machine Learning for Physicists (80.56/100, A), Quantum Mechanics, Electricity and Magnetism, Classical Mechanics, Statistical Physics of Matter

RESEARCH EXPERIENCE

Centre for Polar Observation and Modelling & Dept of Earth Sciences, UCL London, UK
Deep learing for Altimetry Echo Generation 08/2025-06/2026
Research Assistant, Supervisor: Dr. Michel Tsamados
Publication: Deep Learning for Radar Altimetry Echo Generation in Sea Ice Surface Image Recognition(First Author, Under review, IOP: Machine Learning Earth)

- Built a paired Sentinel-3 optical–altimetry dataset over Arctic sea ice by integrating OLCI imagery and SRAL SAR waveforms through temporal-spatial co-location, waveform alignment, and quality-controlled sample construction
- Extracted waveform morphology metrics and retracker-relevant physical descriptors to quantify signal fidelity and downstream geophysical consistency
- Developed a physics-aware cGAN for reconstructing SAR altimetry waveforms from optical observations, and evaluated it using both signal-level and retracker-based physical metrics

Department of Astronomy, Tsinghua University Beijing, China
JWST Data Analysis and Modelling 06/2025-12/2025
Research Assistant, Supervisor: Prof. Zheng Cai

- Utilised 866 JWST multi-band observational data for target selection, image registration, and noise reduction, and constructed a script that can build high quality samples list
- Extracted photometric fluxes and processed 14-band datasets in Python for physical modelling
- Performed Python-based SED fitting on multi-band photometry to estimate stellar mass, age, and star-formation rate

Department of Physics and Astronomy, UCL London, UK
CubeSat Reflectarray Antenna Design (Undergraduate thesis) 01/2025 - 03/2025
Designer, Supervisor: Prof. Giorgio Savini

- Designed and optimized a reflectarray antenna for CubeSats to improve communication efficiency for educational purposes
- Implemented Python-based modelling and conducted Ansys HFSS simulations to analyse and optimize unit cell phase shifts
- Devised a full production pipeline for the reflectarray antenna in a team, ensuring repeatable fabrication
- Validated performance of reflectarray antenna for CubeSat applications

Department of Physics and Astronomy, UCL London, UK

LHC Jet Classification Using Machine Learning 02/2025 - 04/2025

*Model Builder***,** *Supervisor: Prof. Ryan Nichol*

- Constructed a machine learning model to classify jets in the LHC experiment
- Designed a data processing pipeline, applied CNN-based classification, and confirmed the combined "Jet + CALO + Features" model which achieved top performance with 97.04% training accuracy, 0.9978 AUC, and 0.9646 F1 score

Department of Computer Science, Imperial College London WuXi, China

Multi-Label Sentiment Classification Research 07/2024 - 09/2024

Team Leader, Supervisor: Prof. Björn Schulle

Publication: Based on Data Balancing and Model Improvement for Multi-Label Sentiment Classification Performance Enhancement(First Author, Under review, Journal of Information and Telecommunication)

- Developed a sentiment classification model to address dataset class imbalance through the integration of data from GoEmotions, Sentiment140, and GPT-4 mini
- Applied advanced machine learning techniques, including FastText embedding, convolutional layers, BiLSTM, as well as attention mechanisms, to enhance classification performance by 30%
- Achieved high performance on subset metrics: 97% accuracy, 71% precision, 64.5% recall, and a 67.4% F1-score across 28 sentiment categories which surpassed baseline models

WORK EXPERIENCE

(Paid Internship) Centific Technology LTD. (APAC Headquarters) WuXi, China

Data Processing Intern 06/2024 - 09/2024

- Processed backend user data for clients (Microsoft, Apple and Uber) using Pandas from Python
- Developed scripts to convert data into TXT format for manual annotation, improving efficiency by 20% and eliminating manual intervention
- Labelled and processed over 500 data points daily, including text, speech, and video files from backend systems of clients using CVAT and Label Studio
- Conducted quality checks on datasets from external providers, identifying annotation errors and enhancing overall dataset quality by 40%

Shanghai Jingqi Investment Management Co., LTD Shanghai, China

Quantitative Research Intern 07/2023 - 09/2023

- Applied Python to process and analyse stock trading profit and loss data, delivering daily profit and loss predictions for purchased stocks
- Researched systematic quantitative trading strategies in Python, back testing on historical market data, optimising risk-adjusted returns through factor modelling and execution algorithms
- Analysed two datasets, each containing over 100,000 rows, to provide daily profit and loss predictions, including anticipated price fluctuations

SKILLS & LANGUAGES

Languages: English (Advanced), Japanese (Advanced), Mandarin (Native), Cantonese (Native)
Deep Learning: PyTorch (Proficient), TensorFlow; CNNs, conditional GANs, Diffusion models, physics-aware deep learning, loss function design, computer vision, image-to-signal generation
Geophysical Data: SAR remote sensing, SAR altimetry, Sentinel-3 OLCI/SRAL, Sentinel-1,2 data, satellite image processing, radar waveform analysis, sea-ice remote sensing
Scientific Programming: Python (Proficient), MATLAB, NumPy, Pandas, SciPy, xarray, Matplotlib
Annotation & Dataset Tools: CVAT, Label Studio